Rapport PIR ⋆

Ulysse Bourdic-Girard * **Guillaume Claudon** **

* *Ecole Nationale de l'Aviation Civile, Toulouse, France (e-mail: ulysse.bourdic-girard@alumni.enac.fr).*
** *Ecole Nationale de l'Aviation Civile, Toulouse, France (e-mail: guillaume.claudon@alumni.enac.fr)*

Abstract: This study presents a method of leveraging genetic programming principles to determine the flight parameters of a drone swarm controlled by a bio-inspired algorithm developed by Matthieu Verdoucq. The study explores how this method can be used to optimise the behaviour of a drone swarm in a search-and-rescue scenario.

Keywords: Swarm control, Genetic programming

1. INTRODUCTION

Single unmanned aerial vehicles (UAVs) have proven to be exceptionally effective at a wide range of tasks, ranging from cinematography to pesticide application. Swarms of UAVs although not yet commonly used, hold great promise for future applications. One of these is search-and-rescue (SAR). In such scenarios a drone swarm has an evident numerical advantage over a single UAV. Controlling the swarm and ensuring it efficiently covers the search area is however a significant challenge.

Our study is based on the bio-inspired flocking algorithm developed by Verdoucq (2024) . A swarm controlled by this algorithm exhibits emergent behaviours such as schooling (the drones flock together in an orderly fashion), swarming (the swarm remains tight-knit but the drones move erratically), or milling (the drones rotate around swarm's centre in an orderly fashion). Verdoucq et al. (2022) have demonstrated these behaviours emerge from a specific set of parameters which define the equations used to control the drones.

Our goal is to implement genetic programming paradigms to optimise this set of parameters to adapt the swarm's behaviour to a SAR scenario.

2. RELATED WORK

REFERENCES

Verdoucq, M. (2024). *Contrôle bio-inspiré des mouvements collectifs d'un essaim de drones et applications à des scénarios opérationnel.* Ph.D. thesis, Université de Toulouse, Toulouse.

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